Abstract

weakly-

heard and imitated by a Generation 1 participant. The resulting imitated patterns can be given to

experimental

significant predictors of structure (transmission type: $F=7.4$, $p<.01$; generation: $F=14.5$, $p<.001$). Another stepwise model selection suggested that only generation should be entered in the ANOVA as predictor of imitation error. Generation was a significant predictor of imitation error ($F=14.8$, $p<.001$).

We used further ANOVAs to test that these differences were not due to superficial features. We calculated the length of each drumming pattern (in the ‘r space’, see Figure 2), and computed mean and variance of length

88 (s)- (s)8 7. a1uw85 (i)6ae ld 1uoren1uw85 (i12 (a)o6)nea1uu (g)5 6tsi (th6( )a

chain, and 2) the number of patterns in weich each attested motif was discovered at each generation of each chai4o04ooag(0)6(4oner)4hTs04o asgTs04oaler 04ous04oner04oquannerat04oner04o

Figure 4: Normalised histograms for the distribution of interval ratios at each generation in the

e or 9 (i 5 (g) 21 (h) 7 (t) 4 d) 3 ( ) 106 a h i 5 (t) 4 (h) 7 (i 5 n:)] 1 1 0 0 1 0 8 0 0 1 0 8 0 0 1 0 8

The study has several limitations, which could be improved upon with future work. First, we

Acknowledgments

3. Jacoby, N. & J.H. McDermott. 2017. Integer Ratio Priors on Musical Rhythm Revealed Cross-

18. Kirby, S. 2017. Culture and biology in the origins of linguistic structure. *Psychonomic bulletin & review*. **24**

Supplement to: *Evolving building blocks of rhythm: How human cognition creates music via cultural transmission*

initialization:

1: construct an initial representation of interval ratio categories by categorizing, in order, every individual

interval ratio in the initial sequence of 32 *test* pattern7v,ery individeryin the catessentat(32)-18.352.6**0**-2.0000Tdrocedur(the)-2w